

CAPITAL BUDGETING

Capital budgeting is the process of planning and evaluating long-term investment decisions. It involves selecting projects that require large investment and provide benefits over a long period.

The main objective of capital budgeting is to select profitable investment opportunities that maximise the value of the firm and shareholders' wealth.

Objectives of Capital Budgeting

The important objectives are:

1. Maximisation of shareholders' wealth:

Capital budgeting helps select projects that increase the market value of the company.

2. Proper utilisation of funds:

It ensures that limited financial resources are invested in the most profitable projects.

3. Long-term investment planning:

It helps management plan future growth and expansion.

4. Risk evaluation:

It analyses uncertainty associated with future cash flows.

5. Selection of profitable projects:

It helps compare different investment alternatives and choose the best option.

Importance of Capital Budgeting

Capital budgeting is important because:

- Capital investments involve huge amounts of money.
- Decisions affect the company for many years.
- Wrong decisions may cause financial losses.
- Many investment decisions are irreversible.
- It helps control risk and uncertainty.
- It supports business expansion and growth.

1. Payback Period Method

The Payback Period Method is one of the simplest and oldest techniques of capital budgeting. It measures the time required to recover the initial investment made in a project from the cash inflows generated by that project.

In other words, the payback period shows the number of years required for a project to generate enough cash inflows to cover its original cost.

A project is considered more attractive when it recovers its investment in a shorter period because early recovery reduces the risk associated with future cash flows.

If a company invests ₹5,00,000 in a machine and receives annual cash inflows of ₹1,00,000, the payback period will be:

$$\begin{aligned}\text{Payback Period} &= \frac{\text{₹5,00,000}}{\text{₹1,00,000}} \\ &= 5 \text{ Years}\end{aligned}$$

This means the company will recover its original investment after 5 years.

Formula of Payback Period

When annual cash inflows are equal:

$\text{Payback Period} = \frac{\text{Initial Investment}}{\text{Annual Cash Inflow}}$

Where:

- **Initial Investment** = Amount invested at the beginning of the project
- **Annual Cash Inflow** = Cash received every year from the project

Payback Period for Uneven Cash Flows

In many projects, cash inflows are not equal every year. In such cases, cumulative cash inflows are calculated year by year until the initial investment is recovered.

Formula:

$$\text{Payback Period} = \text{Years before full recovery} + \frac{\text{Remaining Investment}}{\text{Cash Flow during Recovery Year}}$$

Example of Uneven Cash Flow

A company invests ₹10,00,000 in a project.

Year	Cash Inflow	Cumulative Cash Flow
1	₹3,00,000	₹3,00,000
2	₹3,50,000	₹6,50,000
3	₹2,50,000	₹9,00,000
4	₹2,00,000	₹11,00,000

Investment recovered during the 4th year.

Remaining investment after 3 years:

$$₹10,00,000 - ₹9,00,000 = ₹1,00,000$$

Payback Period:

$$\begin{aligned} &= 3 + \frac{₹1,00,000}{₹2,00,000} \\ &= 3.5 \text{ Years} \end{aligned}$$

Therefore, the company will recover its investment in **3.5 years**.

Decision Rule of Payback Period

The decision rule is based on the comparison between calculated payback period and the maximum acceptable payback period decided by management.

Accept the Project:

If:

$$\text{Calculated Payback Period} \leq \text{Required Payback Period}$$

The project is accepted.

Reject the Project:

If:

$$\text{Calculated Payback Period} > \text{Required Payback Period}$$

The project is rejected.

When comparing two projects:

- The project with the **shorter payback period** is preferred.
- A shorter recovery period indicates lower investment risk.

Importance of Payback Period Method

1. Helps in Measuring Liquidity

The payback period shows how quickly the invested money can be recovered. Companies facing cash shortages prefer projects with faster recovery.

2. Helps in Risk Assessment

Projects with longer payback periods involve greater uncertainty because future cash flows are difficult to predict. A shorter payback period reduces exposure to future risks.

3. Useful for Initial Screening of Projects

Large organisations often use payback period as a preliminary method to eliminate unsuitable projects before applying advanced techniques.

4. Suitable for Uncertain Business Conditions

When economic conditions are unstable, companies prefer investments that recover funds quickly.

Advantages of Payback Period Method

1. Simple and Easy to Understand

The calculation of payback period is very simple. Even managers without detailed financial knowledge can understand the concept easily.

2. Easy to Calculate

It requires only initial investment and annual cash inflows. Therefore, calculation is less time-consuming compared with discounted cash flow methods.

3. Focuses on Cash Recovery

Unlike accounting-based methods, payback considers actual cash inflows and focuses on recovery of investment.

4. Useful for Risky Projects

Projects with uncertain future benefits can be evaluated using payback because it gives importance to quick recovery.

5. Helps in Liquidity Management

Companies that require quick availability of funds can select projects with shorter payback periods.

Limitations of Payback Period Method

1. Ignores Time Value of Money

The method does not consider the fact that money received today is more valuable than money received in future years. For example, ₹1,00,000 received today is more valuable than ₹1,00,000 received after five years.

2. Ignores Cash Flows After Payback Period

Payback considers only the recovery period and ignores benefits received after the investment is recovered. Example: Two projects may have the same payback period, but one may generate much higher profits after recovery. Payback method cannot identify this difference.

3. Does Not Measure Overall Profitability

The method only measures the recovery period, not the total return earned from the project. A project with a longer payback period may actually be more profitable.

4. No Scientific Basis for Selecting Cut-off Period

The acceptable payback period is decided by management judgment and does not have a fixed financial principle.

5. Not Consistent with Shareholder Wealth Maximisation

The main objective of financial management is to maximise shareholder wealth. Payback does not consider total project profitability or present value of future cash flows.

Payback Reciprocal

Payback reciprocal is another way of expressing the payback period.

Formula:

$$\text{Payback Reciprocal} = \frac{1}{\text{Payback Period}} \times 100$$

It indicates the approximate rate of return generated by a project.

A project with a lower payback period will have a higher payback reciprocal and is considered more attractive.

2. Average Accounting Rate of Return (ARR) Method

The Average Accounting Rate of Return (ARR) is a traditional capital budgeting technique that measures the average annual accounting profit generated by an investment in comparison with the average amount of investment made. ARR is also known as the Accounting Rate of Return or Average Return Method. Unlike the Payback Period Method, which focuses on recovery of investment, ARR focuses on the profitability of the project throughout its entire life.

It uses accounting profits rather than cash flows for evaluating investment proposals.

ARR helps management determine whether the expected accounting return from a project is sufficient compared with the required rate of return.

Formula of ARR

$$\text{ARR} = \frac{\text{Average Annual Profit After Tax}}{\text{Average Investment}} \times 100$$

Where:

Average Annual Profit After Tax:

The average profit earned by the project after deducting depreciation, interest, and taxes during the entire project life.

$$\text{Average Annual Profit} = \frac{\text{Total Profit During Project Life}}{\text{Number of Years}}$$

Average Investment:

$$\text{Average Investment} = \frac{\text{Initial Investment} + \text{Salvage Value}}{2}$$

Where:

- **Initial Investment** = Original cost of the project or asset
- **Salvage Value** = Estimated value of the asset at the end of its useful life

Example of ARR Calculation

A company invests ₹10,00,000 in a machine.

Estimated salvage value after 5 years = ₹2,00,000

Total profit after tax during 5 years = ₹8,00,000

Step 1: Calculate Average Annual Profit

$$\begin{aligned}\text{Average Annual Profit} &= \frac{₹8,00,000}{5} \\ &= ₹1,60,000\end{aligned}$$

Step 2: Calculate Average Investment

$$\begin{aligned}\text{Average Investment} &= \frac{\text{₹}10,00,000 + \text{₹}2,00,000}{2} \\ &= \text{₹}6,00,000\end{aligned}$$

Step 3: Calculate ARR

$$\begin{aligned}\text{ARR} &= \frac{\text{₹}1,60,000}{\text{₹}6,00,000} \times 100 \\ &= 26.67\%\end{aligned}$$

Therefore, the project generates an average accounting return of **26.67%**.

Decision Rule of ARR (Average Accounting Rate of Return)

The decision is made by comparing the calculated ARR with the required rate of return (Target ARR) fixed by management.

Condition	Decision	Explanation
ARR > Required ARR	Accept the Project	The project provides a return higher than the minimum expected return. It indicates better accounting profitability.
ARR < Required ARR	Reject the Project	The project fails to provide the required return expected by management.
ARR = Required ARR	Neutral / Indifferent	The project earns exactly the required rate of return; management may consider other factors before accepting.

When Comparing Different Projects:

Basis	Decision
Higher ARR	Preferred project
Lower ARR	Less preferred project

Basis	Decision
Reason	Higher ARR indicates higher accounting profitability and better expected return.

Importance of ARR Method

1. Measures Project Profitability

ARR shows the average profit earned from an investment during its entire useful life. It helps management understand whether the project generates sufficient accounting returns.

2. Helps in Project Comparison

ARR allows comparison between different investment alternatives. A project with a higher ARR can be selected when other factors remain similar.

3. Uses Accounting Information

ARR uses information available from financial statements such as profit, depreciation and asset values. Therefore, managers can easily calculate and interpret the results.

4. Considers Entire Project Life

Unlike the payback method, ARR considers profits earned throughout the complete life of the project. This provides a broader view of project performance.

5. Useful for Performance Evaluation

ARR can be used to compare actual performance with expected returns after implementation of the project.

Advantages of ARR Method

1. Simple and Easy to Understand

ARR is easy to calculate and interpret. It does not require complex mathematical calculations. Even managers with limited financial knowledge can understand the result.

2. Easy Calculation

The required information for ARR is available from accounting records, making calculation simple and less costly.

3. Considers Profit During Entire Project Life

ARR considers the average profit generated during the complete period of investment. This avoids the limitation of the payback method, which ignores profits after recovery of investment.

4. Useful for Comparing Investment Alternatives

ARR provides a percentage return, making comparison between different projects easier.

Example:

- Project A ARR = 20%
- Project B ARR = 15%

Project A appears more attractive based on accounting return.

5. Recognises Accounting Profitability

Many organisations focus on accounting performance. ARR provides information based on reported profits, which may be useful for internal evaluation.

Limitations of ARR Method

1. Uses Accounting Profit Instead of Cash Flow

The major limitation of ARR is that it uses accounting profit rather than actual cash inflows. Accounting profit includes non-cash items such as depreciation, which may not represent actual financial benefits.

2. Ignores Time Value of Money

ARR does not consider that money received earlier is more valuable than money received later. For example: ₹1,00,000 received today is worth more than ₹1,00,000 received after five years.

3. Depends on Accounting Methods

ARR can vary depending on accounting policies. Different methods of depreciation or asset valuation can change the calculated profit and ARR.

4. Does Not Consider Project Risk

ARR does not adjust returns according to the level of uncertainty involved in the project. A high-return project with high risk and a low-risk project may appear similar under ARR.

5. Ignores Investment Timing

ARR does not consider when profits are generated during the project life. A project generating higher profits in later years may receive the same ARR as a project generating profits earlier.

6. Not Consistent with Wealth Maximisation Objective

The main objective of financial management is maximisation of shareholder wealth. ARR does not consider market value creation, cash flows or present value of future benefits.

3. Net Present Value (NPV) Method

The Net Present Value (NPV) Method is one of the most important and widely accepted discounted cash flow techniques used in capital budgeting. NPV measures the difference between the present value of future cash inflows generated by a project and the present value of cash outflows (investment cost).

The basic principle behind NPV is the time value of money, which states that money available today is more valuable than the same amount received in the future because the current money can be invested to earn returns.

In the NPV method, future cash flows are converted into their present value by applying an appropriate discount rate or cost of capital. The project is accepted if it creates a positive value for the company.

Concept of Present Value

Future cash flows cannot be directly compared with current investment because their values are different due to the time value of money.

Therefore, future cash inflows are discounted back to their current value.

Present Value Formula:

$$PV = \frac{FV}{(1 + r)^n}$$

Where:

- **PV** = Present Value
- **FV** = Future Cash Flow
- **r** = Discount rate / Cost of capital

- **n** = Number of years

A higher discount rate or longer time period reduces the present value of future cash flows.

Definition of NPV

Net Present Value is defined as:

"The difference between the present value of future cash inflows and the present value of cash outflows associated with an investment project."

A positive NPV indicates that the project generates returns higher than the required rate of return.

A negative NPV indicates that the project fails to recover the required return.

Formula of NPV

Basic Formula:

$$\text{NPV} = \text{Present Value of Cash Inflows} - \text{Initial Investment}$$

Example of NPV Calculation

A company invests ₹5,00,000 in a project.

Expected cash inflows:

Year Cash Flow

1 ₹2,00,000

2 ₹2,00,000

3 ₹2,00,000

Discount rate = 10%

Present value of each cash flow is calculated using the discount factor.

Calculate Present Value of Each Cash Flow

Year	Cash Flow	Discount Factor @10%	Present Value
1	₹2,00,000	$1/(1.10)^1 = 0.909$	₹1,81,800
2	₹2,00,000	$1/(1.10)^2 = 0.826$	₹1,65,200
3	₹2,00,000	$1/(1.10)^3 = 0.751$	₹1,50,200

Step 2: Add All Present Values

$$\begin{aligned} &\text{Total PV of Cash Inflows} \\ &= ₹1,81,800 + ₹1,65,200 + ₹1,50,200 \end{aligned}$$

Total Present Value of Cash Inflows = ₹4,97,370

$$\begin{aligned} NPV &= ₹4,97,370 - ₹5,00,000 \\ NPV &= -₹2,630 \end{aligned}$$

Since NPV is negative, the project should be rejected.

Decision Rule of NPV (Net Present Value)

The acceptance or rejection decision is based on the value of **Net Present Value (NPV)**.

NPV Condition	Decision	Reason
NPV > 0 (Positive NPV)	Accept the Project	The project generates returns higher than the required rate of return and creates additional value for shareholders.
NPV < 0 (Negative NPV)	Reject the Project	The project does not generate sufficient returns to recover the investment and fails to create value.
NPV = 0 (Zero NPV)	Project is Indifferent	The project earns exactly the required rate of return; it neither creates nor destroys shareholder wealth.

Importance of NPV Method

1. Considers Time Value of Money

The biggest advantage of NPV is that it recognises the difference between present and future money values. It converts future cash flows into present values before making investment decisions.

2. Uses All Cash Flows During Project Life

Unlike the payback method, NPV considers every cash flow generated throughout the entire life of the project.

It includes:

- Initial investment
- Annual operating cash flows
- Terminal cash flows
- Salvage value

3. Helps in Selecting Mutually Exclusive Projects

When a company has several alternative projects, NPV helps identify the project that creates maximum value. The project with the highest positive NPV is generally selected.

4. Supports Shareholder Wealth Maximisation

NPV directly measures the increase in the value of the firm. A positive NPV project adds value to shareholders.

5. Considers Cost of Capital

NPV uses the company's cost of capital as the discount rate. Therefore, it evaluates whether the project earns more than the required return.

Advantages of NPV Method

1. Scientifically Superior Method

NPV is considered one of the best capital budgeting techniques because it is based on discounted cash flow analysis. It provides a more accurate evaluation compared with traditional methods.

2. Considers Time Value of Money

NPV recognises that future cash flows have lower value compared with current cash flows. This makes investment decisions more realistic.

3. Considers Entire Project Cash Flows

All cash inflows and outflows during the project period are considered. Therefore, it provides a complete picture of project profitability.

4. Consistent with Wealth Maximisation Objective

NPV focuses on value creation rather than only accounting profit. A project with positive NPV increases shareholder wealth.

5. Useful for Comparing Different Projects

NPV allows comparison of different investment alternatives. The project creating the highest value can be selected.

Limitations of NPV Method

1. Complex Calculation

NPV involves discounting multiple future cash flows, which makes calculations more complicated than Payback Period or ARR.

2. Requires Accurate Discount Rate

The accuracy of NPV depends heavily on selecting the correct discount rate. An incorrect discount rate can lead to wrong investment decisions.

3. Difficult for Non-Financial Managers

Managers without financial knowledge may find NPV calculations difficult to understand.

4. Requires Accurate Cash Flow Forecasts

NPV depends on estimated future cash flows. Incorrect assumptions regarding sales, costs or market conditions can affect results.

5. Difficult Comparison in Some Cases

When projects have different investment sizes or different project lives, direct comparison of NPV may sometimes become difficult. It indicates the rate at which both projects have equal NPV.

4. Profitability Index (PI)

The Profitability Index (PI) is a discounted cash flow technique used in capital budgeting to measure the present value of benefits received for every rupee invested in a project. It is also known as the Benefit-Cost Ratio (BCR) or Desirability Factor.

PI shows the relationship between the present value of future cash inflows and the initial investment. It helps management identify how much value is generated from each unit of investment.

For example:

If $PI = 1.20$, it means that for every ₹1 invested, the project generates ₹1.20 in present value benefits.

Formula of Profitability Index (PI)

$$PI = \frac{\text{Present Value of Cash Inflows}}{\text{Present Value of Cash Outflows}}$$

Where:

- **Present Value of Cash Inflows** = Discounted value of future cash receipts
- **Present Value of Cash Outflows** = Initial investment or total investment cost

Alternative Formula:

$$PI = \frac{NPV + \text{Initial Investment}}{\text{Initial Investment}}$$

Example of Profitability Index Calculation

A company invests ₹5,00,000 in a project.

Present Value of future cash inflows = ₹6,00,000

Formula:

$$PI = \frac{\text{PV of Cash Inflows}}{\text{Initial Investment}}$$
$$PI = \frac{₹6,00,000}{₹5,00,000}$$
$$PI = 1.20$$

Interpretation:

PI = 1.20 means:

For every ₹1 invested, the company receives ₹1.20 in present value benefits.

The project creates ₹0.20 additional value per rupee invested.

Decision Rule of Profitability Index (PI)

The acceptance or rejection decision is based on the value of Profitability Index (PI).

PI Condition	Decision	Reason
PI > 1	Accept the Project	The present value of benefits is greater than the investment cost. The project creates additional value for the company.
PI < 1	Reject the Project	The present value of benefits is less than the investment cost. The project results in value loss.
PI = 1	Project is Indifferent	The project earns exactly the required rate of return and neither creates nor destroys value.

Importance of Profitability Index

1. Useful During Capital Rationing

When a company has limited funds but many profitable investment opportunities, PI helps rank projects according to value created per rupee invested.

Projects with higher PI are selected first.

2. Helps Rank Investment Projects

PI provides a common measurement for comparing different projects.

A project with a higher PI is considered more attractive.

Example:

Project PI

A 1.40

B 1.15

C 0.90

Project A will be preferred because it creates the highest value.

3. Shows Value Created Per Rupee Invested

Unlike NPV, which shows total value addition, PI shows the efficiency of investment.

It answers: "How much benefit is generated from each rupee invested?"

4. Considers Time Value of Money

PI uses discounted cash flows, meaning future cash inflows are converted into present values before evaluation.

Advantages of Profitability Index Method

1. Considers Time Value of Money

PI discounts future cash flows and considers that money received in future has lower value compared with current money.

2. Considers All Cash Flows

PI considers cash flows throughout the entire life of the project.

It includes:

- Initial investment
- Annual cash inflows
- Terminal cash flows
- Salvage value

3. Useful When Funds Are Limited

PI is especially useful under capital rationing because it helps select projects that provide maximum return from limited resources.

4. Easy Comparison Between Projects

Since PI is expressed as a ratio, it allows comparison between projects of different sizes.

5. Consistent with Wealth Maximisation

Projects with PI greater than 1 increase shareholder wealth because they create positive value.

Limitations of Profitability Index Method

1. Ranking Problems in Mutually Exclusive Projects

PI may give incorrect ranking when comparing mutually exclusive projects with different investment sizes. Example: A small project may have a higher PI but create less total value compared with a larger project having higher NPV.

2. Requires Accurate Discount Rate

PI calculation depends on the discount rate used. An incorrect discount rate can lead to wrong investment decisions.

3. Difficult Cash Flow Estimation

Future cash flows are estimated values and may differ from actual results due to changes in market conditions.

4. Less Useful Without Capital Constraints

When funds are unlimited, NPV is generally preferred because it directly measures total value addition.

PI vs NPV

Basis	Profitability Index (PI)	NPV
Meaning	Value created per rupee invested	Total value created
Formula	PV Inflows / PV Outflows	PV Inflows - Initial Investment
Result	Ratio	Amount
Use	Capital rationing	Wealth maximisation
Decision	PI > 1 accepted	NPV > 0 accepted

5. Internal Rate of Return (IRR) Method

The Internal Rate of Return (IRR) is an important discounted cash flow technique used in capital budgeting to evaluate the profitability of an investment project. IRR is the discount rate at which the Net Present Value (NPV) of a project becomes zero. In other words, it is the rate of return at which the present value of future cash inflows becomes exactly equal to the initial investment. IRR represents the actual earning capacity or expected return generated by a project.

A project is considered acceptable when its IRR is higher than the required rate of return or cost of capital.

Concept of IRR

Every investment project involves:

- Initial cash outflow (investment)
- Future cash inflows

IRR identifies the maximum rate of return that the project can generate while still recovering the investment.

At IRR:

$$\text{Present Value of Cash Inflows} = \text{Initial Investment}$$

Therefore:

$$\text{NPV} = 0$$

Formula of IRR

Basic Formula:

$$NPV = 0$$

or

$$\text{Initial Investment} = \text{Present Value of Future Cash Flows}$$

2. Interpolation Method

When two discount rates are available:

- One gives positive NPV.
- Another gives negative NPV.

IRR is calculated using:

$$IRR = L + \frac{NPV_L}{NPV_L - NPV_H} \times (H - L)$$

Where:

- **L** = Lower discount rate
- **H** = Higher discount rate
- **NPV_L** = NPV at lower rate
- **NPV_H** = NPV at higher rate

Example of IRR Calculation

A company invests ₹5,00,000 in a project.

Expected future cash inflows generate the following results:

Discount Rate NPV

10% +₹20,000

15% -₹10,000

Since one NPV is positive and another is negative, IRR lies between 10% and 15%.

Using interpolation:

$$\begin{aligned} IRR &= 10 + \frac{20,000}{20,000 - (-10,000)} \times (15 - 10) \\ IRR &= 10 + \frac{20,000}{30,000} \times 5 \\ IRR &= 13.33\% \end{aligned}$$

Therefore, the project's IRR is approximately **13.33%**.

Decision Rule of IRR (Internal Rate of Return)

The acceptance or rejection decision is based on the comparison between IRR and the Cost of Capital (Required Rate of Return).

IRR Condition	Decision	Reason
IRR > Cost of Capital	Accept the Project	The project generates returns higher than the required rate of return. It creates additional value for the company.
IRR < Cost of Capital	Reject the Project	The project does not generate sufficient returns to compensate for the investment.
IRR = Cost of Capital	Project is Indifferent	The project earns exactly the required rate of return; it neither creates nor destroys value.

Importance of IRR Method

1. Measures Actual Earning Capacity of Investment

IRR shows the expected percentage return generated by a project. It helps management understand the profitability level of an investment.

2. Considers Time Value of Money

IRR uses discounted cash flow principles and recognises that future cash flows have lower value compared with present money.

3. Helps Compare Project Returns with Required Return

IRR can be directly compared with the company's cost of capital. If IRR exceeds the required return, the project creates value.

4. Useful for Investment Decisions

IRR helps management select profitable projects and reject projects that do not provide sufficient returns.

Advantages of IRR Method

1. Considers Complete Project Life

IRR considers all cash inflows and outflows during the entire life of the project. It does not ignore cash flows after a certain period.

2. Considers Time Value of Money

Unlike Payback Period and ARR, IRR recognises the importance of timing of cash flows.

3. Easy Comparison with Cost of Capital

IRR is expressed as a percentage, making it easy to compare with required return.

Example:

- Cost of capital = 10%
- IRR = 15%

Since IRR is higher, the project is acceptable.

4. Useful for Ranking Projects

IRR helps compare different investment alternatives based on their expected returns.

5. Consistent with Wealth Maximisation

Projects with IRR higher than the cost of capital generally increase shareholder wealth.

Limitations of IRR Method

1. Difficult Calculation

IRR calculation requires repeated calculations through trial and error or interpolation, making it more complex than Payback Period and ARR.

2. May Produce Multiple IRRs

In projects where cash flows change signs more than once, more than one IRR may exist. Example:

Initial investment → Cash inflows → Additional cash outflow → Cash inflows This creates confusion in decision-making.

3. Assumes Unrealistic Reinvestment Rate

IRR assumes that future cash inflows are reinvested at the same IRR rate.

In practice, reinvestment may occur at the cost of capital, making NPV more reliable.

4. Conflict with NPV Method

For mutually exclusive projects, IRR and NPV may give different rankings due to:

- Difference in project size
- Difference in cash flow timing
- Different project life

When conflict occurs, NPV is generally preferred.

5. Difficult Comparison for Unequal Projects

Projects with different investment sizes or different durations may create misleading IRR comparisons.

IRR vs NPV

Basis	IRR	NPV
Meaning	Rate of return where NPV becomes zero	Value created by project
Result	Percentage (%)	Amount (₹)
Time value	Considered	Considered
Decision	Higher IRR preferred	Higher NPV preferred
Reinvestment assumption	IRR rate	Cost of capital
Reliability	Less reliable in some cases	More reliable

Capital Rationing

Capital Rationing refers to a situation where a company has more profitable investment opportunities than the available funds to finance them. In other words, when a firm has limited capital but several acceptable investment proposals, it cannot accept all projects. Therefore, it must select the best combination of projects that provides maximum return or maximum value creation.

Capital rationing involves the allocation of limited investment funds among competing projects.

Capital rationing can be defined as:

"A situation in which a firm has more acceptable investment proposals than it can finance due to limited availability of funds."

The objective of capital rationing is to select projects that maximise the long-term return of the firm.

Reasons for Capital Rationing

Capital rationing may arise due to two types of factors:

1. External Factors

External capital rationing occurs due to restrictions outside the control of the company.

Examples:

- Limited availability of funds from financial markets.
- Restrictions imposed by banks or financial institutions.
- Imperfect capital markets.
- Government regulations affecting financing availability.

2. Internal Factors

Internal capital rationing occurs due to management decisions.

Reasons include:

- Management may avoid raising additional external funds.
- Fear of increasing financial risk.
- Avoiding dilution of ownership control.
- Limited managerial resources to handle many projects.
- Maintaining a specific debt-equity ratio.

Types of Capital Rationing Situations

Capital rationing problems are mainly divided into two situations:

Divisible Projects with Single Period Constraint

In divisible projects, investment can be divided into smaller portions.

Steps:

1. Calculate the **Profitability Index (PI)** of each project.

$$PI = \frac{PV \text{ of Cash Inflows}}{Initial Investment}$$

2. Rank projects according to PI.
3. Select projects starting from the highest PI until the available funds are fully utilised.

Example:

Project	Investment	PI	Ranking
A	₹2,00,000	1.50	1
B	₹3,00,000	1.30	2
C	₹2,00,000	1.10	3

Project A will be selected first because it provides the highest value per rupee invested.

Indivisible Projects with Single Period Constraint

In indivisible projects, the entire project must be accepted or rejected. Partial investment is not possible.

Steps:

1. Prepare all possible combinations of projects.
2. Calculate the total investment of each combination.
3. Calculate the total NPV of each combination.
4. Select the combination with the highest NPV within the available budget.

Importance of Capital Rationing

1. Efficient Allocation of Limited Funds

Capital rationing helps companies invest available funds in the most profitable projects.

2. Maximises Shareholder Wealth

By selecting projects with higher returns, companies can increase shareholder value.

3. Helps Project Ranking

It provides a systematic method for ranking investment proposals.

4. Controls Excessive Investment

It prevents companies from investing beyond their financial capacity.

5. Improves Financial Planning

It helps management plan investment decisions according to available resources.

Advantages of Capital Rationing

Advantage	Explanation
Better fund utilisation	Ensures limited funds are invested in profitable projects.
Risk control	Prevents excessive borrowing and financial pressure.
Project selection	Helps choose the best combination of investments.
Wealth maximisation	Supports selection of value-creating projects.

Limitations of Capital Rationing

Limitation	Explanation
May reject profitable projects	Some good projects may be ignored due to limited funds.
Difficult estimation	Requires accurate calculation of cash flows and returns.
Depends on ranking method	Different methods may produce different project rankings.
Artificial restriction	Internal capital limits may prevent profitable expansion.